

Electron-Ion Collider, Brookhaven National Laboratory			
Doc No. EIC-PSG-SPC-005	Author: Dominic Prevosto	Effective Date: [Insert here]	Review Frequency: N/A
Specification: Technical Specification for the EIC Rack Enclosures			Revision: 00

TECHNICAL SPECIFICATION
for the
EIC Rack Enclosures

Quality Classification: A-3

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REVISION HISTORY

Revision #	Effective Date	Summary of Change
00		Initial Release

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LIST OF ACRONYMS

ASME	American Society of Mechanical Engineers
BNL	Brookhaven National Laboratory
BSA	Brookhaven Science Associates
EIC	Electron-Ion Collider
ESH	Environment, Safety and Health
HX	Heat Exchanger
RF	Radio Frequency

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EIC Rack Enclosures

1 SCOPE

This specification defines the technical requirements for EIC Rack Enclosures to be used at the Electron-Ion Collider (EIC) at Brookhaven National Laboratory (BNL) managed by Brookhaven Science Associates (BSA).

1.1 Background

The EIC Project will design, build, and install the accelerator hardware for a new electron storage ring, an electron injector, and modifications to the hadron ring required to produce an electron-ion collider and infrastructure. It will include one interaction region, with an allowance for a second, and one detector. The facility is designed to meet requirements to ensure high reliability and availability for the user program.

The rack enclosures will be housing many different devices for a variety of groups at BNL. These enclosures will be holding power supply equipment, RF equipment, controls equipment, and so forth.

2 APPLICABLE DOCUMENTS

The following specifications, standards and drawings form a part of this specification to the extent specified herein. Unless otherwise specified, the applicable revision level of these documents shall be that in effect on the date of the solicitation or contract. Exceptions must be approved in writing by BSA.

In the event of a conflict between the applicable documents and this specification, the Contractor shall immediately notify the BSA Contractual Representative who shall in each instance determine which document takes precedence and advise the Contractor accordingly. Failure to notify BSA of a document conflict shall not relieve the Contractor's responsibility to ensure full compliance to all requirements.

Document Number	Document Title
NEMA 250 Type 12	Enclosures for Electrical Equipment (1000 Volts Maximum)
UL 50 / UL 50E	Enclosures for Electrical Equipment

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3 REQUIREMENTS

BSA is considering 2 different cooling methodologies for varying applications required to support the EIC project.

Fan cooled ambient air. This is a rack with no heat exchanger, where fans direct cool filtered, ambient air from outside into the cabinet, and/or hot air from inside the rack to the outside.

Heat Exchanger (HX) Cooled. This is a sealed rack with a water-to-air heat exchanger inside of it which circulates air in a closed loop. The general design is that the HX blows cool air to the front plenum created by the front door, sides, and chassis front panels. Air flows from the front through the chassis (or between chassis) to the heat exchanger return.

3.1 Technical/Performance Characteristics

3.1.1 Equipment Racks General Characteristics

Racks will have varying overall dimensions as specified herein. Racks will also have differing penetration requirements, depending on the cooling method, and cable access needs. These penetrations will either have blank panels, fans, filters, or other modifications to allow for cable access, AC delivery, or water fittings. The rack Contractor shall have the capability to accommodate to this level of modularity.

The cooling system shall be designed such that cooling is delivered and heat is removed to maintain even temperature throughout the loaded rack.

The racks shall be designed and constructed such that condensation does not form on the enclosures. Insulation shall be included if required.

All rack components shall be electrically bonded together. Grounding studs and a grounding busbar shall be provided.

All enclosure panels including doors, sidewalls, and top sides shall be directly connected to the frame to provide proper electrical grounding capability.

Racks shall have removable side panels.

Mounting provisions shall allow for multiple racks to be bolted together in a side-by-side fashion.

Racks shall have gasketed front and rear doors.

Front door shall contain a large viewing window that allows visibility to all rack slots.

The internal racks shall have standard "U" rails configured to accept 19-inch equipment via 10-32 tapped holes in standard U spacing.

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All rack solutions shall have adjustable front, middle and rear rails.

All rack structural components and sheet metal assemblies shall be designed with sufficient flexibility to accommodate updates and modifications based on BSA requirements.

Door latches shall be key lockable, but capable of being opened without using a key when unlocked.

Racks shall have lifting points on the top of the rack to allow the attachment of rated hoist rings and shall be sufficient to accommodate lifting and transport in fully loaded conditions.

Racks shall be delivered on a temporary fork liftable transport platform.

Racks shall have leveling feet to accommodate floor adjustments.

3.1.2 Rack Cooling Configurations and Specifications

3.1.2.1 Fan Cooled Ambient Air-Cooling Specifications

The racks shall have filters to allow ambient air into the cabinets.

Cabinets shall have necessary filtering provisions to allow for clean continuous operation.

The racks shall have an opening at the front bottom of the cabinet for cool ambient air entry.

The heat exchanger shall have a blower or impeller at the front, which directs the air up along the front door of the cabinet.

The blower or impeller shall be capable of providing a flow rate sufficient to meet the target interior temperature based on the provided ambient air.

The racks shall have means to allow for warm air to exit the rear of the rack mounted chassis units, out to the top rear of the cabinet.

The cabinet air flow design shall be capable of removing up to 3kW of heat.

The rack enclosure shall conform to one of the following dimensional configurations, as required for the specified cooling arrangement:

- 52U, 31.5" Width, 42" Depth.
- 52U, 22" Width, 36" Depth.

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Specifications:

- Max Supply air temperature of 78°F
- Stability Requirement of $(75 \pm 3^\circ\text{F})$
- Max combined BSA rack equipment heat dissipation of 3kW
- Max combined BSA rack equipment air flow of 900 cfm

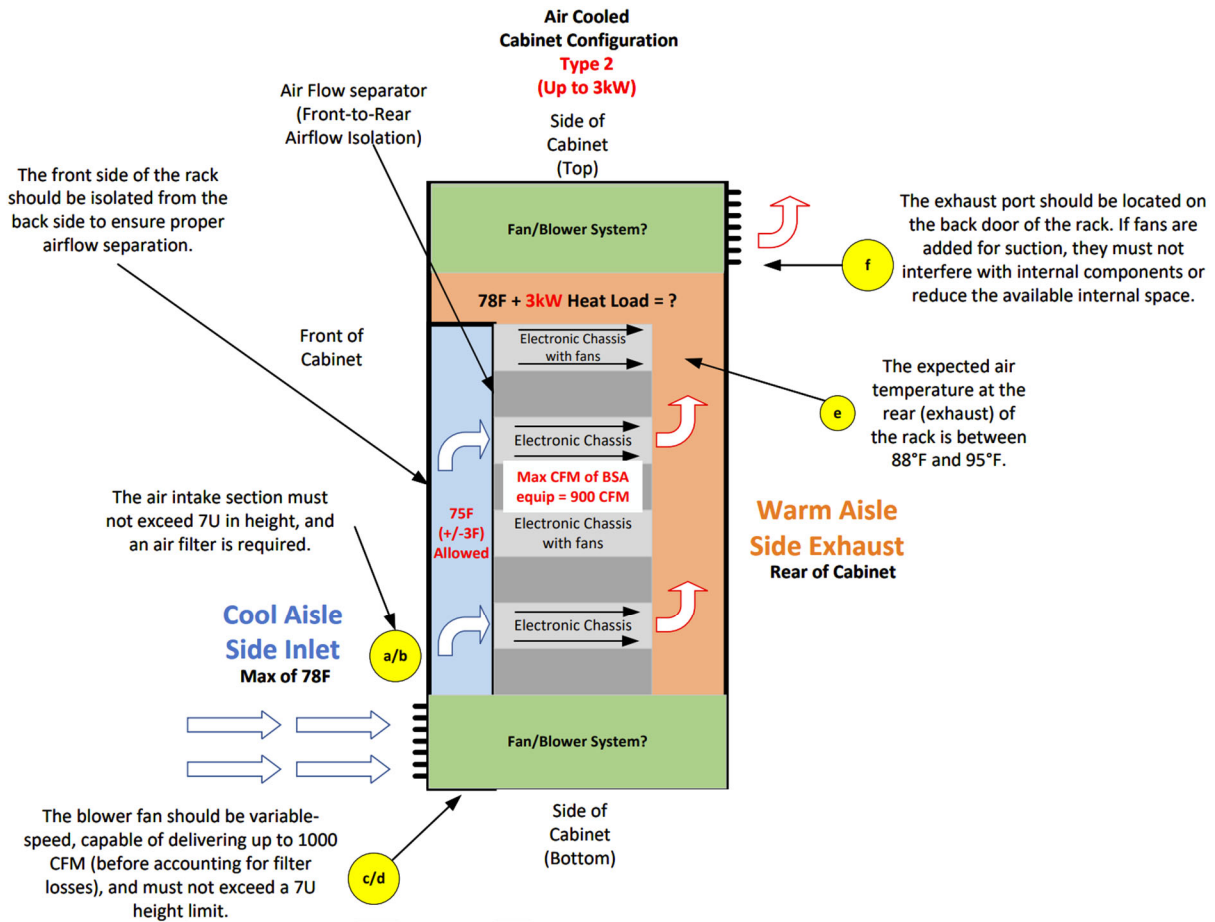


Figure 2: Fan Cooled Configuration Type 2

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3.1.2.2 Heat Exchanger Cooled Design Specifications

The heat exchanger shall be water-cooled.

The water-cooled heat exchanger shall meet the standards for NEMA12 performance per UL50 at the interface between the heat exchanger and the rack group being cooled. It shall be UL Listed, labeled, and marked as such.

The heat exchanger shall be capable of removing up to 8kW of heat depending on the configuration.

The heat exchanger cabinet solution shall have an integrated temperature sensor that provides feedback to the cooling source for dynamic adjustments.

The heat exchanger shall have a blower or impeller at the front, which directs the air up along the front door of the cabinet.

The heat exchanger shall be capable of providing a flow rate sufficient to meet the interior temperature based on its size.

Rack mounted equipment will pull the cool air through the chassis and exhaust the warm air to the rear of the cabinet.

The heat exchanger shall pull the warm air down and re-circulate the conditioned air to the front of the cabinet.

The water supply and return connection fittings shall be supplied as the last connection before entering the rack for easy disconnection and access inside the rack.

Supply and return shall have manual shutoff valves for service.

Any water control valves shall be inboard to the rack enclosure.

All racks shall be fully gasketed for tight, leak-proof installation in compliance with NEMA12 Enclosure Rating and UL 50 and marked as such.

The door construction shall be in compliance with NEMA 12 standards, meeting deflection and resistance to dust and dripping liquids requirements per the UL 50 Standard.

The rack structure shall be capable of supporting a minimum of 400kg of BSA furnished equipment.

The dimensions for this configuration are 52U, 31.5" Width, 42" Depth.

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Specifications:

- Max internal air supply temp to BSA equipment of 75.5°F
- Stability Requirement of $(75 \pm 0.5^\circ\text{F})$
- Max combined BSA rack equipment heat dissipation of 8kW
- Max combined BSA rack equipment air flow of 900 cfm

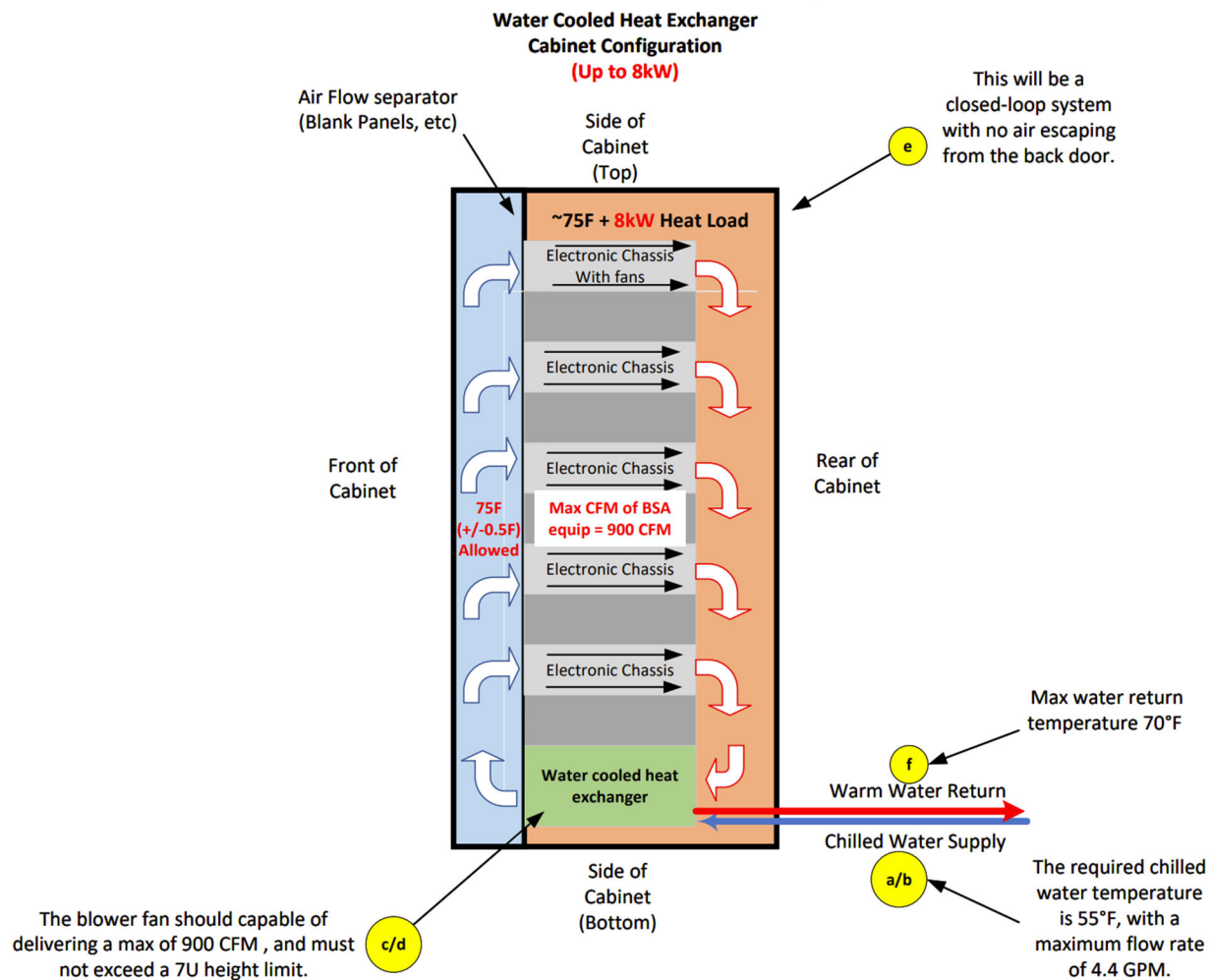


Figure 3: Water Cooled Heat Exchanger Configuration

3.2 Code/Standard Compliance

- NFPA 70 National Electric Code (Latest Edition)
- UL 1995 Heating and Cooling Equipment
- NEMA 250 Type 12

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- UL 50 Cabinets and Boxes
- NRTL Listed

3.3 Environment, Safety and Health (ESH)

Identification of hazards, and their mitigation, shall occur during all stages of the life of the Equipment including; installation, commissioning, operation, maintenance, and repair. Identification of means to eliminate or manage each hazard shall be provided based on the following hierarchy:

- Eliminate the hazard, or manage the hazards in the following hierarchy;
- Substitute with something less hazardous, or
- Engineer – mitigate the severity of the hazard, or
- User Personal Protective Equipment (least desirable).

3.4 Marking and Serialization

A stainless-steel type "C" nameplate shall be mounted in plain sight, giving all standard information, including the following:

- Manufacturer's Name and Address.
- Manufacturer's Part Number.
- Manufacturer's Type and Serial Number.
- Gross Weight of the Unit.
- Date of Manufacture.

3.5 Packaging/Shipping

The Contractor shall be responsible for packaging, handling, and shipment of all rack enclosures to ensure safe delivery without damage to any mechanical, electrical, or cosmetic components.

All rack enclosures shall be securely packaged to protect against shock, vibration, moisture, dust, and environmental exposure during transportation and storage. Packaging methods shall be appropriate for the size and weight of the assemblies and shall include, at a minimum:

- Protective wrapping and internal bracing to prevent movement during transit
- Shock-absorbing materials where required
- Protection of all external surfaces

Each rack enclosure shall be delivered mounted on a temporary, forklift-compatible transport base suitable for safe loading, unloading, and staging. The Contractor shall coordinate with the BSA

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Technical Representative to implement a delivery schedule, aligned with production progress and site readiness.

4 VERIFICATION

Appropriate inspection/tests shall be performed to ensure Equipment conforms to all requirements in section 3 of this specification.

4.1.1 Preproduction Tests (First Article)

Prior to production of the full quantity of rack enclosures, the Contractor shall provide one (1) of each fully assembled rack enclosure type for First Article inspection and testing. The First Article unit shall be used to verify compliance with all mechanical, structural, electrical grounding, and airflow requirements specified in this document. The following tests and inspections shall be performed.

4.2 Inspection/Testing Details/Requirements

The rack enclosures shall undergo a detailed visual inspection to verify compliance with all drawings and mechanical specifications.

The inspection shall verify:

- Overall rack dimensions and tolerances
- Mounting rail spacing and alignment
- Proper installation of all hardware, fasteners, and structural members
- Quality of welds, joints, and mechanical connections
- Surface finish, paint, and coating quality
- Door alignment and operation (front and rear)
- Proper installation of side panels and removable sections
- Absence of sharp edges, deformation, or cosmetic defects

Representative equipment (or simulated loads) shall be installed into the rack to verify:

- Proper alignment with mounting rails
- Adequate clearance for installation and removal
- Compatibility with standard rack mounting hardware
- Adequate access for cable routing and maintenance

Since the racks will contain high power supplies, airflow validation is important.

Verification shall include:

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- Confirmation of vent locations and sizes
- Verification that airflow paths are not obstructed by structural components
- Validation that equipment airflow is not restricted

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